



SENTINEL GUARD

USER MANUAL

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1. Introduction

The Sentinel Guard system is a complete physical and technological IoT security monitoring solution that provides a secure method of collecting, transmitting, and displaying real-time temperature and humidity sensor data on a cloud-based dashboard. The Sentinel Guard system and its cloud-based dashboard are built with a security-first design to ensure data integrity and security at every step in the data's journey from the physical IoT device to the cloud.

This manual will discuss overall operation of the Sentinel Guard system, how to operate the cloud-based dashboard, and how to interpret sensor readings and alerts received from the cloud.

1.1 What sentinel Guard Does

Feature	Description
Collects environmental data	A small hardware sensor reads temperature and humidity from the environment every few seconds.
Transmits data securely	Data is encrypted and digitally signed before being sent, ensuring it cannot be read or altered in transit.
Detects threats automatically	The system checks incoming data packet for signs of tampering, flooding, false data, and replay attack attempts before accepting it.
Displays everything live	A cloud dashboard shows current sensor readings, historical logs, and any security alerts in real time.

1.2 Glossary of Key Terms

Term	Definition
IoT Device	Any small electronic device that gathers and/or transmits information (e.g., smart sensor or thermostat) connected to the Internet.
Sensor Node	The physical equipment (ESP32 board and DHT22 sensor) which collects the temperature and humidity and sends it to the system.
Encryption	Data scrambling, allowing for data to be read by only the intended recipient.
Digital Signature (HMAC)	An identifier code on each data packet that verifies the integrity of the data, or that the data has not been altered since it was sent.
TLS	A security protocol that creates an encrypted tunnel for data travelling across the internet.
MQTT Broker	A relay service to receive data packets from the sensor and to relay them to the server.
Dashboard	The web interface for managing the system, viewing alerts and monitoring live sensor data.
Firestore	Cloud database for storing sensor readings and security events that are being monitored by Sentinel Guard on Google's behalf.
Replay Attack	An attack in which an already captured legitimate data packet is replayed to the system to cause it to recognize old data as new.

DDoS Attack	An attack that involves sending so much traffic in that the system becomes overloaded and crashes.
False Data Injection	An attack in which values of sensors are fabricated or manipulated to deceive the system.
Anomaly	A measurement taken by a sensor that is very different from and out of the normal range

2. System Overview

Sentinel Guard is composed of five data processing stages working together as a pipeline with respect to its responsibility to fulfil a defined operation either through security or process before sending the information on to another stage in the pipeline.

2.1 Data flow

The following lists the flow of data from sensor to dashboard:

- **ESP32 Sensor** – The ESP32 sensor records and encrypts the temperature/humidity with digital signatures and publishes the encrypted data payload to broker.
- **Mosquitto MQTT Broker** – The MQTT broker accepts the encrypted data payload from the ESP32 broker and authenticates to determine whether the sender has valid credentials and then forwards the accepted encrypted payload to the server.
- **Server** – The python receiver will check the digital signature of the packet, decrypt the packet, perform all security checks, and process the data using the machine learning models.
- **Cloud Database** – Any readings that have been accepted as valid will be posted to Firebase Firestore along with any security events that are detected while processing the data during the previous three steps.
- **Dashboard** – The web dashboard pulls data from Firestore to show live sensor values, alarm notifications, and historical logs.

Note: Any packet that fails a security check is rejected, logged as a security event, and never stored as valid sensor data.

2.2 System Components

Component	Role
Sensor (ESP32 + DHT22)	Reads temperature and humidity. Encrypts and signs data before transmission.
MQTT Broker (Mosquitto)	Secure message relay. Enforces authentication and rejects unauthorized connections.
Python Server	Processes every packet through a security verification pipeline and runs the anomaly detection model.
Firebase Firestore	Cloud database storing all sensor readings, system logs, and security events.
ML Anomaly Detector	A trained Machine Learning model that classifies each reading as normal or anomalous.
Web Dashboard	Browser-based interface for monitoring data and alters, with role-based access control (RBAC).

3. Getting Started

Access to the system is via the web-based dashboard in a typical browser, after it has been configured and running by your administrator.

3.1. Requirements

- An Internet connected device (computer, tablet, or cell phone)
- A Google account enrolled by your administrator with the system.
- The URL for your dashboard as given by your administrator.

3.2. Logging In

- open the Dashboard URL in your browser
- Tap on Sign in with Google
- Click your registered Google account and finish sign in
- You'll end up on the Live Sensor Data page

3.3. User Roles

Every user is assigned one of two roles by the administrator,

Role	Permissions
Admin	Full access: All devices and sensor data, user accounts & roles, all security events & audit logs, acknowledge and dismiss alerts.
Standard User	View data of sensors assigned to users, and view alerts related to them. No user management permissions and only read only access.

4. Dashboard Guide

The dashboard has four main sections, accessible from the navigation menu.

4.1. Live Sensor Data

Shows the latest value of the sensor, refreshing periodically every few seconds.

Field	Description
Temperature	The temperature displayed in Celsius degrees.
Humidity	The current relative humidity in percentage.
Packet Status	This is the overall status of packet data. To check if the data sent to the DHT 22 sensor by the temperature and humidity readings are being received completely, partially or not at all.
Anomaly Detection	The machine learning model's classification. Normal will indicate the reading is within expected patterns, and Anomaly Detected will indicate the reading is outside of expected patterns.

Confidence Score	The amount of confidence the model has in the classification and a percentage.
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- If readings are not updating for several minutes, the sensor device could have stopped. Contact your administrator.

4.2. Alerts and Security Events

Displays a list of all the security events detected by the system, in chronological order. Every entry contains the alert Type, the date and time detected and a short description.

Alert Type	What it means	Recommended Action
Tampered Packet	Digital signature received on a packet did not match, data was altered since sent.	Rejected automatically. If this alert occurs often, please inform your administrator.
Replay Attack	An attacker resends a packet that has been previously captured.	Blocked automatically. If you continue to notice it, inform your administrator.
DDoS Attack	System detected excessive amount of packets received.	A cooldown block will automatically activate. Monitor and notify your administrator if it continues.
False Data Injection	The values received from the sensor were outside the range that is physically possible.	Rejected automatically. Notify your administrator.
Unencrypted Packet	A packet received unencrypted, which means that someone is communicating directly with the broker, bypassing the sensor.	Let your administrator know right away.

Anomaly Detected	A reading was identified as statistically unusual by the machine learning model.	Analyze the trends of temperature and humidity in the recent past. If there is no apparent environmental cause, contact your administrator.
Sequence Mismatch	There was sophisticated tampering indicated by the internal inconsistencies in the packet's numbering system.	Rejected automatically. If it occurs more than once, let your administrator know.
Processing Error	Unable to process or understand the packet.	Contact the administrator to check system logs.

4.3. Historical Logs

Ensures complete, cloud based historical log of all sensor readings and security events, searchable. This page lets you review previous readings, look up particular types of events or display trends over time.

4.4. User Management (Admin Only)

Administrators can add, remove, and manage user accounts from this page

- Click “Add a new user” and enter the email address of the Google account you want to add
- Set and/or modify a user's role (Admin or Standard User)
- To revoke a user of the dashboard, click Remove a user.

Don't give Grant Admin access to any other users than trusted team members. Standard User is suitable for the majority of users.

5. Security Features

Sentinel Guard applies multiple independent security layers at different points in the data pipeline. Each layer defends against a specific type of threat. If one layer is bypassed, the others continue to provide protection.

5.1 Data Encryption (AES-128)

Before transmitting any data, the sensor encrypts the reading using AES-128, a widely used and trusted encryption standard. The data is scrambled into an unreadable format that can only be unlocked by the server, which holds the matching key. Anyone intercepting the data in transit will see only meaningless encrypted content.

5.2 Digital Signature (HMAC-SHA256)

Each data packet carries a unique digital fingerprint generated from the encrypted content. When the server receives the packet, it independently recalculates this fingerprint. If the two match, the data is genuine. If they differ even by a single character the packet is immediately rejected. This ensures that any tampering in transit is detected before the data is ever decrypted.

5.3 Encrypted Connection (TLS)

The entire communication channel between the sensor and the broker is protected by TLS, the same protocol that secures online banking and web browsing. This creates an encrypted tunnel that prevents anyone on the network from reading or interfering with the connection.

5.4 Authentication

The broker requires a valid username and password before accepting any connection. Devices or applications that cannot provide the correct credentials are blocked from connecting entirely.

5.5 Replay Attack Protection

Every packet includes an incrementing sequence number. The server tracks the last accepted sequence number and rejects any packet with a number that has already been processed. If five or more replay attempts are detected within ten seconds, the device is temporarily blocked for thirty seconds and the event is logged.

5.6 DDoS Protection

If the system receives more than twenty packets within a five-second window, it identifies the traffic as a flood attack and activates a ten-second cooldown, during which all incoming packets are blocked. The broker also enforces limits on packet size and queue depth, preventing flood traffic from reaching the server in the first place.

5.7 False Data Detection

Every accepted packet is checked against the physical operating limits of the DHT22 sensor. Any temperature or humidity value that falls outside the possible range for that sensor is immediately rejected. This catches extreme injection attempts before the machine learning model is even invoked.

5.8 Machine Learning Anomaly Detection

After passing all rule-based checks, each reading is analysed by a trained machine learning model. The model has learned from historical data what normal sensor behaviour looks like, and it can flag readings that appear unusual even when they fall within physically valid limits. Flagged readings are saved with an anomaly label, and an alert is raised on the dashboard.

The full security verification pipeline, in order, is:

- Receive → DDoS Check → Structure Validation → Digital Signature Verification → Replay Check → Decryption → Sequence Validation → Range Check → ML Detection → Cloud Upload

6. Normal Operation

When the system is running as expected, the dashboard will show the following:

Indicator	What It Means
Readings update every few seconds	The temperature and humidity values on the Live Data page refresh automatically.
Packet Status: Valid	Every accepted reading has passed all functionality checks.
Anomaly Detection: Normal	The machine learning model is satisfied that readings fall within expected patterns.
No new security alerts	The Alerts page shows no recent activity.

6.1 Starting the System

Starting the system is handled by the administrator. The components must be started in a specific order: the broker first, then the server, then the sensor. As a dashboard user, you simply wait for readings to appear before beginning to monitor.

6.2 Stopping the System

Stopping the system is also handled by the administrator. All data that has already been received and uploaded to Firestore remains accessible in the Historical Logs even after the system is stopped.

7. Troubleshooting

Problem	Likely Cause	Action
No sensor data on the dashboard	The sensor or server may not be running	Confirm with admin that all components are active and connected. Refresh the page after a minute
Unable to login	Your account may not be registered, or the wrong account is being used	Contact admin to verify your account has been added
Live data stopped updating	The sensor, broker, or server may have stopped unexpectedly	Notify your admin. Historical data will still be accessible in the logs
Large number of security alerts	The system may be under attack, or a hardware fault	Do not dismiss the alerts. Review the alert types and notify the admin with the details
Anomaly detected appears frequently	Could indicate a real environmental change or a subtle attack pattern	Check whether there is a physical explanation. If not notify the admin
Dashboard page not loading	A browser or network issue	Refresh the page, clear your browser cache, or try a different browser

8. Conclusion and future work

Sentinel Guard demonstrates a complete, production based IoT security monitoring system that addresses real world security challenges affecting modern IoT deployments. Combination of hardware sensor nodes, encrypted communications, server-side cryptographic verification, machine learning classification and a real-time cloud dashboard, the system provides strongly layered defense against common IoT attack vectors.

The defense-in-depth architecture ensures that no single layer failure can compromise the system. Each security layer functions independently, and bypassing one layer does not expose the system to threats defended by other layers.

Planned Improvements

Improvement	Description
Cloud hosted broker	Deploy the MQTT broker to a cloud server to enable remote monitoring beyond the local network.
TLS 1.3	Upgrade to TLS 1.3 for stronger cipher suits and improved handshake performance when ESP32 library support becomes available.
Automated key rotation	Periodically rotate AES and HMAC keys automatically without requiring firmware updates.
Mobile push notifications	Deliver real-time security alerts to mobile devices via Firebase Cloud Messaging
Multi-device supports	Extend the system to manage multiple sensor nodes with per-device dashboards and access control.
Advanced ML model	Explore LSTM-based models for time-series anomaly detection with improved accuracy on sequential sensor data
Mutual TLS (mTLS)	Issue client certificates to each sensor device for mutual authentication at the broker level.

Appendix – A

Alert	Meaning	Action
Tampered Packet	Digital signature check failed. Data was modified in transit	Rejected automatically. Notify admin if its frequent.
Replay attack	A previously sent packet was resent.	Blocked automatically. Notify admin if persistent
DDoS attack	Packet flood detected above threshold	Cooldown activated automatically. Notify admin if it continues
False data injection	Sensor values outside physically possible range	Rejected automatically. Notify admin
Unencrypted Packet	Unencrypted data sent directly to the broker	Serious. Notify admin immediately
Anomaly detected	ML model flagged the reading as statistically unusual.	Review trends. Notify admin if unexplained.
Sequence mismatch	Internal packet numbering was inconsistent	Rejected automatically. Notify admin if repeated.